

Name **Suggested Solutions**



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Class	Reg Number
17	

Calculator Model / No.

Chemistry

9729/01

Paper 1

21 September 2018
1 hour

Additional Material: *Data Booklet*
OMS

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces provided at the top of this page.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the use of OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of OMR Answer Sheet

Ensure you have written your name, class register number and class on the OMR Answer Sheet.

Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **OMR sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

This document consists of **13** printed pages (including this cover page)

Qn	Ans
1	B
2	C
3	A
4	A
5	D
6	B
7	B
8	C
9	C
10	D

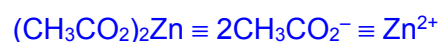
Qn	Ans
11	D
12	B
13	D
14	A
15	A
16	A
17	D
18	D
19	B
20	D

Qn	Ans
21	B
22	B
23	D
24	A
25	C
26	B
27	C
28	C
29	A
30	C

1 Answer: B

$$[(\text{CH}_3\text{CO}_2)_2\text{Zn}] = \frac{10.64}{183.4} = 0.0580 \text{ mol dm}^{-3}$$

$$\text{Amount of } (\text{CH}_3\text{CO}_2)_2\text{Zn} = \frac{5}{1000} \times 0.0580 = 2.90 \times 10^{-4} \text{ mol}$$



$$\text{Total amount of ions} = 3 \times 2.90 \times 10^{-4} = 8.70 \times 10^{-4} \text{ mol}$$

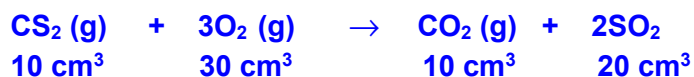
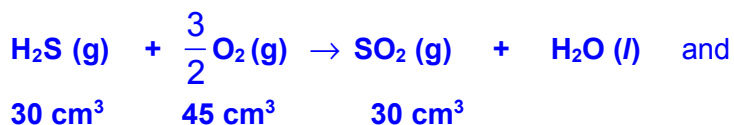
$$\text{Total number of ions} = 8.70 \times 10^{-4} \times 6.02 \times 10^{23} = \underline{5.2 \times 10^{20} \text{ ions}}$$

2 Answer: C

Explanation:

$$\text{volume of H}_2\text{S} = \frac{3}{4} \times 40 = \underline{30 \text{ cm}^3} \text{ and volume of CS}_2 = \frac{1}{4} \times 40 = \underline{10 \text{ cm}^3}$$

Combustion equations:



$$\text{Total volume of SO}_2 \text{ formed} = 30 + 20 = \underline{50 \text{ cm}^3}$$

$$\text{Volume of CO}_2 \text{ formed} = \underline{10 \text{ cm}^3}$$

$$\text{volume of O}_2 \text{ remaining} = 100 - (45 + 30) = \underline{25 \text{ cm}^3}$$

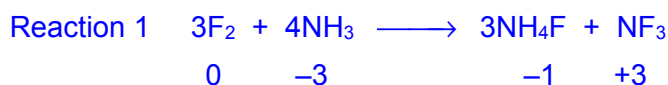
$$\text{Volume of gaseous mixture after burning} = 50 + 10 + 25 = \underline{85 \text{ cm}^3}$$

Since CO₂ and SO₂ are acidic gases, they will be removed by NaOH (aq)

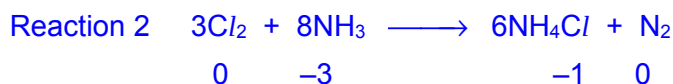
$$\text{therefore, Volume of gaseous mixture after adding NaOH (aq)} = 85 - 60 = \underline{25 \text{ cm}^3}$$

3 **Answer: A**

By considering change in oxidation number of element:



NH₃ acts as reducing agent hence it is a stronger reducing agent than F₂.



NH₃ acts as reducing agent hence it is a stronger reducing agent than Cl₂.

F₂ oxidises N to a larger extent (from -3 in NH₃ to +3 in NF₃) as compared to Cl₂ (from -3 in NH₃ to 0 in N₂). **F₂ is a stronger oxidising agent and hence a weaker reducing agent as compared to Cl₂.**

NH₃ is the **strongest reducing agent** whereas F₂ is the **weakest reducing agent**.

4 **Answer: A**

For element **L**: biggest increase between 6th and 7th ionisation energy.

(Largest difference in IE between 6th and 7th I.E.)

7th electron is removed from the inner quantum shell which is closer to the nucleus.

Thus the element has 6 valence electrons.

Element **L** belongs to **Group 16**.

For element **M**: biggest increase between 3rd and 4th ionisation energy.

(Largest difference in IE between 3rd and 4th I.E.)

4th electron is removed from the inner quantum shell which is closer to the nucleus.

Thus the element has 3 valence electrons.

Element **M** belongs to **Group 13**.

So the likely formula of the compound formed is **L₃M₂**.

5 Answer: D

	X	Y
Proton number, n	n	n + 1
No of electrons	n + 1	n
Charge	+1	+2

Option A is wrong

X and **Y** have different number of protons. Hence they can't be isotopes.

Option B is wrong

X and **Y** have different number of protons. **X** atom has n electrons while **Y** has (n + 1) electrons. Hence **X** and **Y** atoms have different full electronic configurations.

Option C is wrong

Y has charge of +2 while it has same number of electrons as that of **X**. Hence, **Y** has smaller ionic radius. This leads to higher charge density.

Option D is correct

Y has charge of +2 while it has same number of electrons as that of **X**. Hence, **Y** has smaller ionic radius.

6 Answer: B

	First molecule	Second molecule
A	CO_2 (2 bond pairs) Shape: Linear Polarity: Non-polar Bond angle: 180°	F_2O (2 bond pairs, 2 lone pair) Shape: Bent Polarity: Polar Bond angle: 104.5°
B	NF_3 (3 bond pairs, 1 lone pair) Shape: Trigonal Pyramidal Polarity: Polar Bond angle: 107°	SO_3 (3 bond pairs, 0 lone pair) Shape: Trigonal Planar Polarity: Non-polar Bond angle: 120°
C	SF_6 (6 bond pairs, 0 lone pair) Shape: Octahedral Polarity: Non-polar Bond angle: 90°	XeF_4 (4 bond pairs, 2 lone pair) Shape: Square Planar Polarity: Non-polar Bond angle: 90°
D	SO_2 (2 bond pairs, 1 lone pair) Shape: Bent Polarity: Polar Bond angle: 118°	NO_2 (2 bond pairs, 1 lone electron) Shape: Bent Polarity: Polar Bond angle: $> 118^\circ$
	Lone pair – bond pair repulsion > 1 unpaired electron – bond pair repulsion.	

7 **Answer: B**

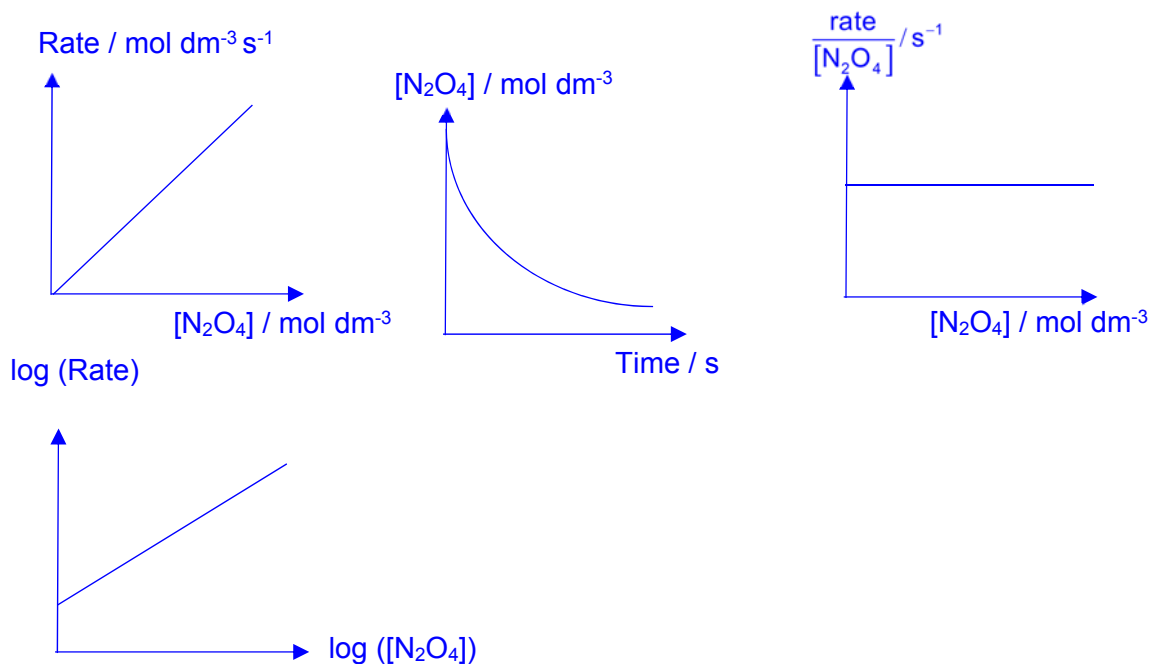
Statement 1 is correct as melting involves overcoming of the strong metallic bonds arising from the strong electrostatic forces attraction between the delocalised electrons and the residual metallic cations.

Statement 2 is correct as Al contribute more delocalised electrons than Na and this contributes to stronger metallic bonds.

Statement 3 is incorrect as electrical conductivity in metals is due to the delocalised electrons, not mobile ions.

8 **Answer: C**

The graphs for first order reaction are shown as below:

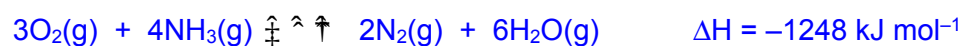
9 **Answer: C**

$$\text{Half-life, } t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{\ln 2}{0.09} = 7.7 \text{ h}$$

$$\frac{C_t}{C_o} = \left(\frac{1}{2}\right)^n$$

$$\frac{18}{100} = \left(\frac{1}{2}\right)^n \Rightarrow n = 2.47$$

$$\text{Time taken} = \text{half-life} \times n = 7.7 \times n = \underline{\underline{19.0 \text{ h}}}$$

10 Answer: D

When pressure increases, by Le Chatelier's Principle, the equilibrium position shifts left to reduce the number of moles of gas to decrease pressure.

Therefore, Q represents products since the graphs show decreasing %Q when pressure increases.

Since forward reaction is exothermic, $T_2 > T_1$. By Le Chatelier's Principle, the equilibrium position shifts left towards the endothermic reaction to absorb heat.
 T_2 has lower %Q than T_1

11 Answer: D

Option 1 is correct as it involves the formation of an intermediate.

Option 2 is correct as the potential energy of the intermediate is lower than that of the starting material.

Option 3 is correct as the catalyst will lower both E_a and E_a'

12 Answer: B

When temperature increases, based on Le Chatelier's principle, equilibrium position will shift right towards the endothermic reaction to absorb heat.

$[\text{H}^+]$ & $[\text{OH}^-]$ increases in magnitude

Since $K_w = [\text{H}^+][\text{OH}^-]$, K_w increases in magnitude.

$[\text{H}^+] = [\text{OH}^-]$ as 1 mol of water molecule dissociates to form 1 mol of H^+ and 1 mol of OH^- .

pH will decrease to a value below 7, as $[\text{H}^+]$ increases. ($\text{pH} = -\log [\text{H}^+]$)

13 Answer: D

Alkaline buffer consists of a weak base and its conjugate acid.

$$\text{pOH} = \text{pK}_b + \lg \frac{[\text{salt}]}{[\text{acid}]}$$

Maximum buffer capacity occurs when $\text{pOH} = \text{pK}_b$, where $\frac{[\text{salt}]}{[\text{acid}]} = 1$

A $\rightarrow$ 0.003 mol NH_4^+ and 0.001 mol of NH_3

B $\rightarrow$ acidic buffer

C $\rightarrow$ 0.02 mol of $\text{CH}_3\text{CH}_2\text{NH}_3^+$

D $\rightarrow$ 0.0025 mol CH_3NH_3^+ and 0.0025 mol of CH_3NH_2

14 Answer: A

Bromophenol-blue appears purple ($\text{pH} > 4.6$) $\Rightarrow$ Option **A** is **incorrect**

Phenol-red appears yellow ($\text{pH} < 6.8$) $\Rightarrow$ Options **C** and **D** are **incorrect**.

The solution must be weakly acidic and with $4.6 < \text{pH} < 6.8$.

15 Answer: A

The calculation could be completed using ΔH°_f of all compounds if ΔH°_f of $\text{Cl}\cdot$ is provided. Atomisation of $\text{Cl}-\text{Cl}$ molecule involves homolysis of the bond, forming Cl atoms with one unpaired electron each, i.e. Cl radicals.

16 Answer: A

$$\Delta G = \Delta H - T\Delta S$$

Reaction is exothermic $\Rightarrow \Delta H < 0 \Rightarrow$ y-intercept is negative

Less gaseous products than reactants $\Rightarrow \Delta S < 0 \Rightarrow (-\Delta S) > 0 \Rightarrow$ gradient of graph is positive

17 Answer: D

Explanation:

BaCl_2

$$\begin{aligned} \Delta H^\circ_{\text{solution}} &= |\text{Lattice energy}| - |\Delta H^\circ_{\text{hydration}} \text{Ba}^{2+}| - 2|\Delta H^\circ_{\text{hydration}} \text{Cl}^-| \\ &= 2053 - 1309 - 2(378) = -12 \text{ kJ mol}^{-1} \end{aligned}$$

18 **Answer: D**

Option A is incorrect.

Cathode is a positive electrode and accept electrons.

Option B is incorrect.

$$E_{\text{cell}} = E_{\text{red}} - E_{\text{ox}} = 0.54 - (-3.04) = +3.58 \text{ V}$$

Option C is incorrect.

Ni is a metal which allows **electrons** to pass through and connect Li to I₂ complex.

Option D is correct.

$$\text{Amount of Li}^+, \frac{5}{6.9} = \frac{0.8 \times 10^{-3} \times t}{1 \times 96500} \rightarrow t = 8.73 \times 10^7 \text{ s} \rightarrow \text{Time} = 2.77 \text{ years}$$

19 **Answer: B**

$$E_{\text{red}}^{\ominus} = E_{\text{Cr}^{3+} | \text{Cr}^{2+}}^{\ominus} = -0.41 \text{ V} \qquad E_{\text{ox}}^{\ominus} = E_{\text{Fe}^{2+} | \text{Fe}}^{\ominus} = -0.44 \text{ V}$$

$$E_{\text{cell}}^{\ominus} = E_{\text{red}}^{\ominus} - E_{\text{ox}}^{\ominus} = -0.41 - (-0.44) = \underline{+0.03 \text{ V}}$$

For E_{cell} to be 0.00 V, E_{red} should decrease (more negative) **OR**
 E_{ox} should increase (more positive)

Change	Effect
increase [FeSO ₄]	increase in [Fe ²⁺ (aq)], Fe ²⁺ + 2e ⁻ ⇌ Fe eqm position shifts right, <u>E_{ox} ↑</u>
decrease [CrCl ₂]	decrease in [Cr ²⁺ (aq)], Cr ³⁺ + e ⁻ ⇌ Cr ²⁺ eqm position shifts right, <u>E_{red} ↑</u>
decrease [CrCl ₃]	decrease in [Cr ³⁺ (aq)], Cr ³⁺ + e ⁻ ⇌ Cr ²⁺ eqm position shifts right, <u>E_{red} ↓</u>
increase surface area of Fe electrode	no effect to E_{red} value

20 **Answer: D**

Option **A** is wrong. Step 1 is reduction.

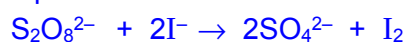
Option **B** is inconclusive. While the relative strength of H_2O , SCN^- and EDTA^{4-} ligands can be concluded from the ligand exchange reactions in Step 4 and 5, nothing can be concluded about the relative strength of H_2O and CN^- ligands.

Option **C** is incorrect. While $\text{Fe}(\text{OH})_3$ can be formed, CO_2 (g) cannot be produced from this reaction. The correct reagent to be used in Step 2 should be Na_2CO_3 .

Option **D** is correct.

21 **Answer: B**

Option 1 is correct.



$$E^\ominus_{(\text{S}_2\text{O}_8^{2-} / \text{SO}_4^{2-})} = +2.01\text{V} ; E^\ominus_{(\text{I}_2 / \text{I}^-)} = +0.54\text{V}$$

Hence, any transition metal ions catalyst with **$E^\ominus$ values between + 2.01 V and +0.54 V**

$$E^\ominus_{(\text{VO}_2^+ / \text{VO}^{2+})} = +1.00\text{V}$$

Option 2 is incorrect.

When $\text{VO}\text{SO}_4(\text{H}_2\text{O})_5$ dissolve in water to form an aqueous solution. $[\text{VO}(\text{H}_2\text{O})_6]^{2+}$ is formed. The blue colour of the solution observed is due to the d-d transition caused by the splitting of the d-orbitals by the water ligands.

Option 3 is correct.

Zinc metal is functioning as a reducing agent. Assuming Zn is in excess,

$E^\ominus_{(\text{Zn}^{2+} / \text{Zn})} = -0.76\text{V}$ (Oxidation) $E^\ominus_{(\text{VO}^{2+} / \text{V}^{2+})} = +0.34\text{V}$ (Reduction)	$\text{Zn} + 2\text{VO}^{2+} + 4\text{H}^+ \rightarrow \text{Zn}^{2+} + 2\text{V}^{3+} + 2\text{H}_2\text{O}$ $E^\ominus_{\text{cell}} = +0.34 - (-0.76) = +1.10\text{V} > 0$ (feasible)
$E^\ominus_{(\text{Zn}^{2+} / \text{Zn})} = -0.76\text{V}$ (Oxidation) $E^\ominus_{(\text{V}^{3+} / \text{V}^{2+})} = -0.26\text{V}$ (Reduction)	$\text{Zn} + 2\text{V}^{3+} \rightarrow \text{Zn}^{2+} + 2\text{V}^{2+}$ $E^\ominus_{\text{cell}} = -0.26 - (-0.76) = +0.50\text{V} > 0$ (feasible)
$E^\ominus_{(\text{Zn}^{2+} / \text{Zn})} = -0.76\text{V}$ (Oxidation) $E^\ominus_{(\text{V}^{2+} / \text{V})} = -1.20\text{V}$ (Reduction)	$E^\ominus_{\text{cell}} = -1.20 - (-0.76) = -0.55\text{V} < 0$ (not feasible)

22 Answer: B

1	The oxide of E is amphoteric.	E is aluminum .
2	The oxide of F dissolves in water to form a strongly alkaline solution.	F is sodium .
3	The oxide of G reacts with dilute sodium hydroxide at room temperature.	G could either be phosphorus or sulfur .
4	The oxide of H is insoluble in water but is soluble in concentrated sodium hydroxide.	H is silicon .

In increasing proton number,
sodium, aluminium, silicon, phosphorus/sulfur

Option **B**: **F, E, H, G**

23 Answer: D

Element **A** has the **highest 3rd IE** $\Rightarrow$ element **A** is in Group 2.

A : Mg Group 2, **B : Al** Group 13,
C : Si Group 14, **E : S** Group 16

- 1 Mg is a strong reducing agent ($E^{\ominus}_{\text{Mg}^{2+}/\text{Mg}} = -2.38\text{V}$). (False)
- 2 Al_2O_3 does not dissolve in water, because of its extremely high lattice energy. Large amount of energy is required to break the strong ionic bonds. (False)
- 3 SiO_2 is an acidic oxide, even though it doesn't dissolve in water to form an acid. It reacts with hot conc. NaOH to form Na_2SiO_3 . (True)
- 4 Sulfur is in Group 16, hence it can form a hydride of H_2S . (True)

24 Answer: A

Option 1 – Comparing BaCO_3 and CaCO_3 :

Ionic radius: Ba^{2+} (0.135 nm) > Ca^{2+} (0.099 nm)

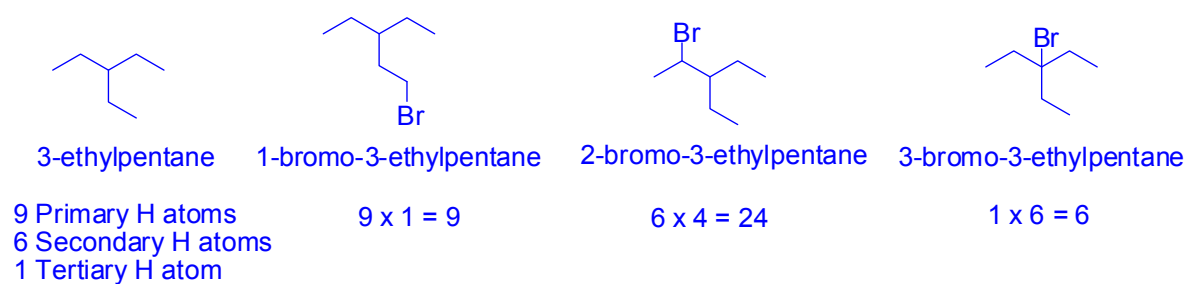
$$|\text{Lattice energy}| \propto \frac{q_+q_-}{r_+ + r_-}$$

Magnitude of lattice energy: $\text{CaCO}_3 > \text{BaCO}_3$

Option 2 & 3 – Comparing BaCO_3 , CuCO_3 and CaCO_3 :

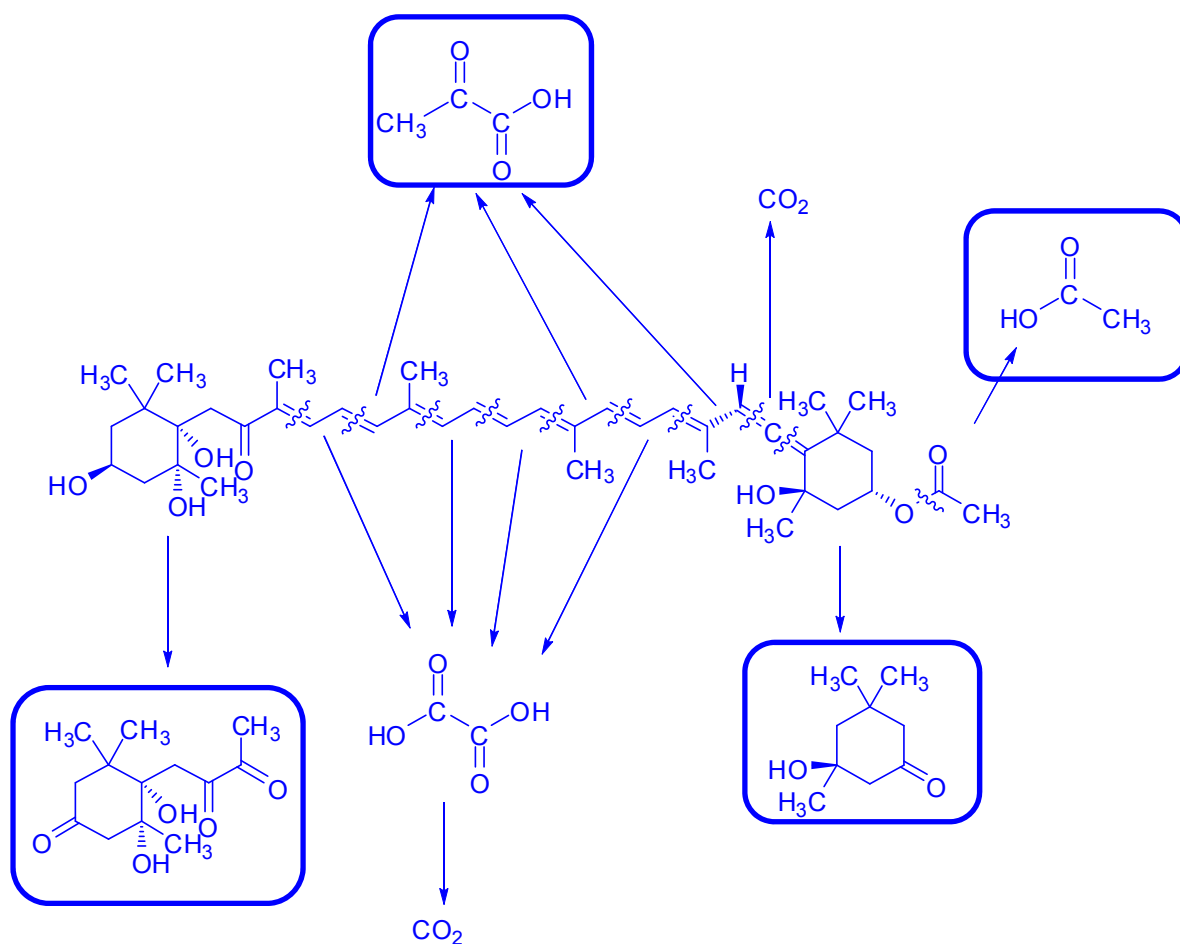
- Ionic radius: Cu^{2+} (0.073 nm) < Ca^{2+} (0.099 nm) < Ba^{2+} (0.135 nm)
- Charge density & polarising power of cation: $\text{Cu}^{2+} > \text{Ca}^{2+} > \text{Ba}^{2+}$
- Ability of cation to distort the CO_3^{2-} electron cloud, weakening and break the C–O bond for: $\text{Cu}^{2+} > \text{Ca}^{2+} > \text{Ba}^{2+}$
- Thermal Stability: $\text{BaCO}_3 > \text{CaCO}_3 > \text{CuCO}_3$

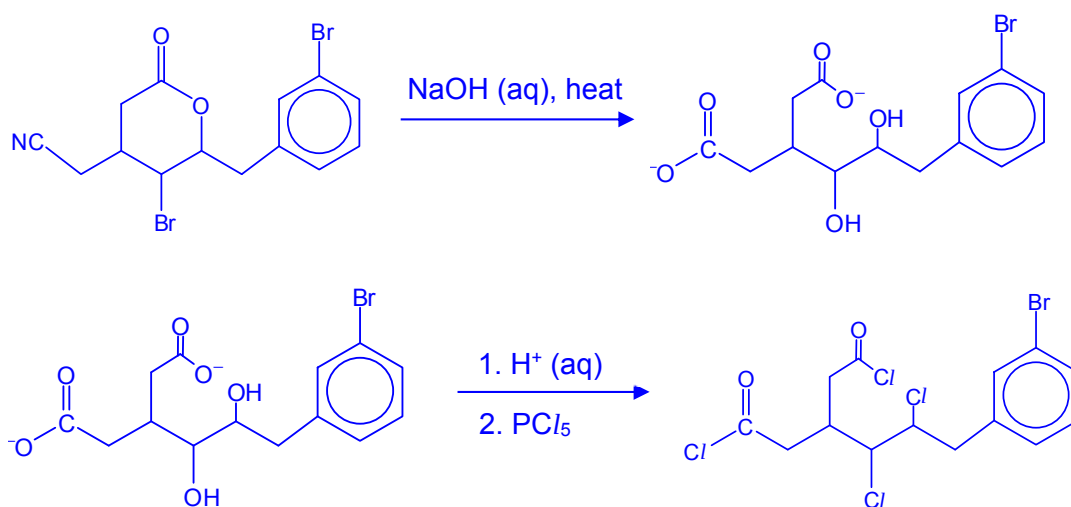
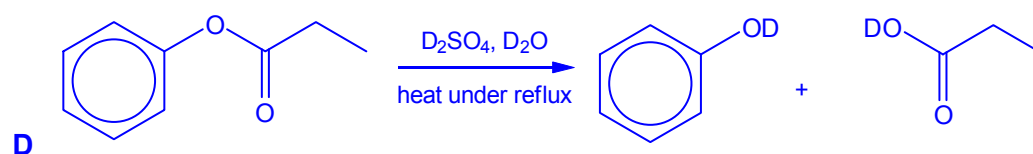
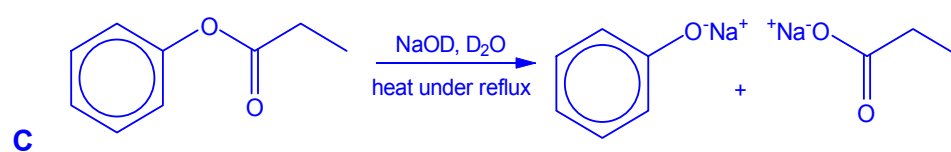
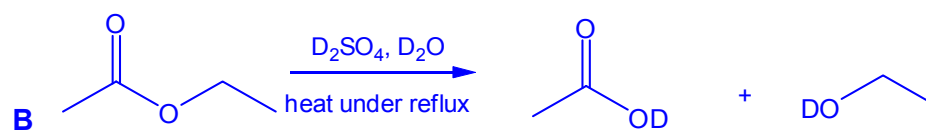
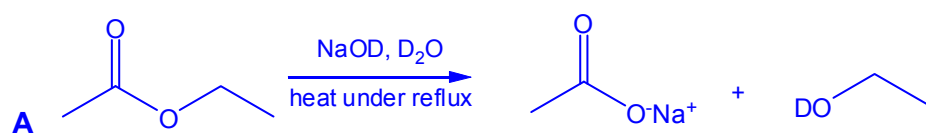
25 **Answer: C**

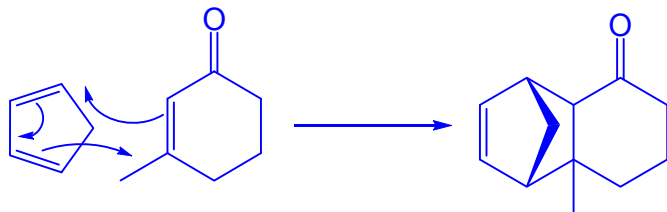


Ratio is 9 : 24 : 6
= 3 : 8 : 2

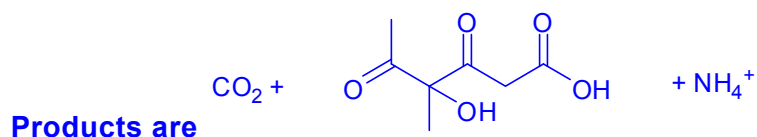
26 **Answer: B**



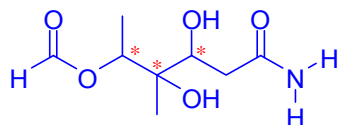
27 **Answer: C**28 **Answer: C**

29 **Answer: A****Explanation:**30 **Answer: C****Explanation:**

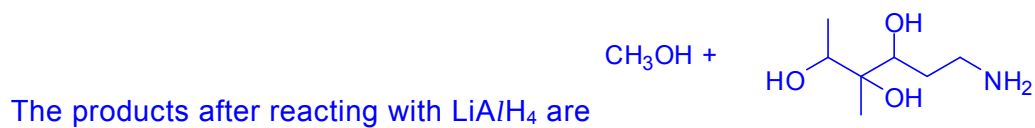
Option 1 is true: Compound A undergoes acidic hydrolysis (amide and ester) as well as oxidation.



Option 2 is false: Compound A has 3 chiral carbon atoms with no internal plane of symmetry. Total number of stereoisomers should be $2^3 = 8$ (not $2 \times 3 = 6$)



Option 3 is false:

**END OF PAPER**